

ENTERPRISE MILESTONE SPECIFICATION

Module 7: Human Resource Management, Biometric Attendance, Bahrain Labour Law Leave, Automated Payroll & Gratuity / EOSB

Comprehensive milestone definition, resource allocation matrix, 16-week delivery roadmap, and structured commercial payment governance for the enterprise workforce platform of Popular Auto Spare & A/C Parts Co. W.L.L.

MODULE IDENTIFIER

SL-POP-ERP-MS-007

BA REFERENCE DOCUMENT

DOC-007 v1.0 (04-July-2026)

TOTAL ENGINEERING EFFORT

16 Working Weeks (4.0 Months)

FIXED PROJECT COST

BD 6,094.548

TARGET ARCHITECTURE

Enterprise HRMS, Biometric Sync & WPS Payroll

DOCUMENT STATUS

FINAL PROPOSAL

Section 1.0

Executive Summary & Master ERP Context

Strategic Alignment

The **Human Resource Management & Biometric Payroll System** (Module 7) represents the workforce operations, statutory compliance, and payroll automation cornerstone for Popular Auto Spare & A/C Parts Co. W.L.L. As an expanding enterprise operating multi-branch retail outlets, central distribution warehouses, and logistics fleets across the Kingdom of Bahrain, managing employee lifecycle records, synchronizing biometric attendance clocks, automating complex shift rosters, enforcing Bahrain Labour Law (Law No. 36 of 2012) leave entitlements, and generating Central Bank WPS salary files are paramount for statutory compliance and employee satisfaction.



Core Objective & Business Purpose

To establish an automated, fully compliant human resource and workforce operations ecosystem consolidating Employee 360 Dossiers, multi-branch Biometric Attendance synchronization, Bahrain Labour Law statutory leave accruals, WPS Central Bank payroll bank files, GOSI calculations, and Article 116 End of Service Indemnity (EOSB) governance.

MASTER ERP TRANSFORMATION ARCHITECTURE (9 CORE
PROCESS MODULES)

CURRENT SCOPE: MODULE 07
ACTIVE

01 PCode Gen & Item Master

02 Vendor & Purchase Flow

03 Stock Verification & Location

04 Sales & POS Checkout

05 Finance, Tax & VAT Accounting

06 Administration & Security

07 HRMS & Biometric Payroll

08 Hardware & Barcode Infrastructure

09 Executive BI & Mobile Analytics



Employee 360 & Expiry Radar

Centralized staff profiles, encrypted document repository, LMRA/CPR/Passport expiry tracking, and proactive renewal alerts.

Employee 360

LMRA / CPR Alerts

Document Dossier



Biometric Attendance & Leave Rules

Real-time ZKTeco biometric clock sync, multi-shift roster allocation, and Bahrain Labour Law 30-day leave accrual engine.

Biometric Sync

Shift Roster

30-Day Leave



WPS Payroll, GOSI & EOSB Gratuity

Automated gross-to-net salary processing, Central Bank of Bahrain WPS bank file export, GOSI tables, and Article 116 indemnity.

CBB WPS File

GOSI Engine

Art. 116 EOSB

Section
2.0

Module 7: Functional Scope & Architecture Breakdown (BA Ref: DOC-007)

Full Specification
Mapping

Derived directly from the 80 pages of **DOC-007: Human Resource Management & Payroll Business Analysis Report**, the implementation scope is structured across 6 core operational pillars:

2.1 Employee 360 Master, Digital Dossier & Expiry Governance

- **Comprehensive Profile Registry:** Personal details, emergency contacts, nationality, passport numbers, CPR smart-card records, blood group, and educational records.
- **Statutory Document Vault:** Secure encrypted storage for digital copies of Passports, Visas, LMRA Work Permits, Bahrain Driving Licenses, and Employment Contracts.
- **Proactive Expiry Radar:** Automated 90, 60, and 30-day notifications for Passport, Visa, CPR, and Driving License expirations with HR dashboard flags.
- **Employment History & Designations:** Department allocation, reporting hierarchy, job grading, probation tracking, and internal promotion/transfer records.
- **Bank & WPS Configuration:** Employee IBAN registry, bank code, and salary distribution preferences configured for Central Bank of Bahrain WPS formatting.

2.2 Biometric Hardware Integration, Shift Rostering & Overtime Rules

- **Hardware Daemon Synchronization:** Background service pulling punch logs from multi-branch ZKTeco biometric time-clocks via TCP/IP and HTTP push protocols.
- **Flexible Shift Roster Engine:** Configuration of single-shift, split-shift, and rotational warehouse/counter shifts with customizable grace periods.
- **Automated Lateness & Early Exit Penalties:** Configurable deduction rules according to corporate HR policy and Bahrain Labour Law parameters.
- **Ramadan Working Hours Adjustment:** Automated reduced work schedules (6-hour daily caps) for Muslim employees during the Holy Month of Ramadan.
- **Overtime Engine:** Automated calculation of normal overtime (1.25x / 125%) and rest day / public holiday overtime (1.50x / 150%) linked directly to monthly payroll.

2.3 Bahrain Labour Law Leave Management & Accrual Engine

- **Statutory Annual Leave Accrual:** Automated monthly accrual of 2.5 working days per completed month (30 calendar days per year) as per Law No. 36 of 2012.
- **Sick Leave Tier Configuration:** 15 days full pay, 20 days half pay, and remaining days without pay with medical certificate upload requirements.
- **Special Statutory Leaves:** Hajj leave (14 days paid), marriage leave (3 days paid), bereavement leave (3 days paid), and maternity/paternity allowances.
- **Leave Encashment Engine:** Automatic valuation of unutilized annual leave balances based on basic salary plus standard allowances upon employee exit or request.
- **Public Holiday Calendar:** Multi-year Bahrain statutory holiday scheduler with automated shift roster overrides and overtime adjustments.

2.4 Automated Payroll Processing, CBB WPS Export & GOSI Compliance

- **Gross-to-Net Computation:** Automated salary processing factoring basic pay, housing, transport, phone, overtime, unpaid leave deductions, and loan repayments.
- **Bahrain WPS Bank File Generator:** 100% compliant SIF/WPS electronic transfer file formatted to Central Bank of Bahrain specifications for direct bank upload.
- **GOSI Statutory Engine:** Automated computation of Bahraini national contributions (8% employee / 12% employer) and expatriate contributions (1% employee / 3% employer).
- **Salary Advances & Staff Loans:** Multi-installment loan repayment tracking with monthly automated payroll deduction scheduling.
- **Finance GL Payroll Integration:** Automated generation and posting of journal vouchers to the General Ledger (Module 5) across expense and liability accounts.

🔑 2.5 End of Service Benefits (EOSB) & Statutory Indemnity (Article 116)

- **Statutory Gratuity Engine:** Strict implementation of Bahrain Labour Law Art. 116 (15 days basic wage per year for first 3 years, 1 month per year thereafter).
- **Resignation vs. Termination Rules:** Scaled entitlement logic based on completed years of continuous service according to statutory resignation guidelines.
- **Comprehensive Final Settlement Slip:** Automated consolidation of final prorated salary, leave balance encashment, gratuity entitlement, and deduction of company dues.
- **Clearance Checklist Workflow:** Multi-department sign-off protocol (IT assets, tools, vehicle return, finance advances, locker key return) before final settlement release.

📱 2.6 Employee Self-Service (ESS) Portal & Mobile Requests

- **Responsive Mobile ESS Portal:** Clean, smartphone-friendly interface allowing staff to view personal dossiers, attendance logs, and leave balances.
- **Digital Leave & Loan Applications:** Online request submissions with multi-level manager approval workflows and real-time status tracking.
- **Encrypted Payslip Downloads:** Password-protected monthly payslip generation with detailed earning and deduction breakdowns.
- **HR Document Requests:** Automated issuance requests for salary certificates, embassy NOC letters, and bank letters with dynamic PDF generation.

Section 3.0

Project Team Structure & Dedicated Resource Pricing (16 Weeks / 4.0 Months)

Standard
Rate Card

In strict accordance with the **SaNDS Lab Dedicated Engineering Rate Card**, the resource allocation and cost breakdown for the 16-week (4.0 months / 80 working days) delivery of Module 7 is structured as follows (all figures in Bahraini Dinars - BHD):

ENGINEERING ROLE	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H/MO)	EFFORT ALLOCATION	TOTAL COST (BHD)
Project Manager & Enterprise Solutions Architect	BD 545.455	BD 136.364	BD 3.409	4.0 Months (16 Wks)	BD 2,181.820
Back-End Lead Engineer (PHP MVC / HRMS Engine)	BD 227.273	BD 56.818	BD 1.420	4.0 Months (16 Wks)	BD 909.092
Front-End Lead Engineer (UI/UX / ESS Portal)	BD 227.273	BD 56.818	BD 1.420	4.0 Months (16 Wks)	BD 909.092
Database & Cloud Infrastructure Architect	BD 204.545	BD 51.136	BD 1.278	4.0 Months (16 Wks)	BD 818.180
QA & System Test Automation Lead Engineer	BD 159.091	BD 39.773	BD 0.994	4.0 Months (16 Wks)	BD 636.364
Dedicated Biometric Hardware & Network Integrator	BD 159.091	BD 39.773	BD 0.994	4.0 Months (16 Wks)	BD 636.364
Total Dedicated Engineering Team	BD 1,523.636 / mo	BD 380.909 / wk	BD 9.518 / hr	16 Weeks (4.0 Mo)	BD 6,094.548



Fixed Price Guarantee

Total project investment for Module 7 is fixed at **BD 6,094.548**. No hidden deployment, integration, or hourly surcharges apply within the agreed functional scope.

Section 4.0

Detailed Milestone Breakdown & Deliverables (16-Week Roadmap)

Sprint Execution
Plan

The implementation is organized into 4 sequential 4-week milestones (25.00% each), guaranteeing focused execution, continuous integration, and transparent deliverable acceptance:

1

Milestone 7.1: Employee 360 Master, Document Dossier & Biometric Synchronization

Weeks 1 - 4
(4 Wks)

Architecture and database setup of Employee 360 profiles, digital document repository, automated expiry alerts for CPR/Passports/Visas, and real-time biometric time-clock daemon synchronization across all branch locations.

✓ Employee Master & Dossier:

Comprehensive personal, contact, bank, and passport database with encrypted file attachments.

✓ Biometric Sync Daemon: Automated

TCP/IP service pulling live punch data from multi-branch ZKTeco clocks.

- ✓ **Expiry Alert Engine:** 90/60/30-day proactive notification triggers for CPR, Visa, Passport, and driving licenses.

2

Milestone 7.2: Shift Rostering, Overtime Rules & Bahrain Labour Law Leave Accruals

Weeks 5 - 8
(4 Wks)

Implementation of flexible shift rosters, split-shift configurations, automated lateness/early exit penalties, 125%/150% overtime rules, and Bahrain Labour Law 30-day annual leave accrual ledger.

- ✓ **Shift & Roster Engine:** Multi-branch shift calendars, grace period rules, and Ramadan hour adjustments.
- ✓ **Bahrain Leave Engine:** 2.5 days/month annual leave accrual, sick leave tiering, and public holiday overrides.
- ✓ **Overtime Matrix:** Automatic calculation of normal overtime (1.25x) and holiday/weekend overtime (1.50x).

3

Milestone 7.3: Automated Payroll Engine, CBB WPS File Generator & GOSI Calculation

Weeks 9 - 12
(4 Wks)

Engineering of automated gross-to-net salary engine factoring allowances, deductions, unpaid leave, loan installments, Central Bank of Bahrain WPS bank export, and GOSI contribution schedules.

- ✓ **Gross-to-Net Payroll:** Automated calculation with transparent breakdown of earnings, deductions, and loan installments.
- ✓ **CBB WPS File Export:** 100% compliant SIF/WPS electronic file generator for direct upload to Bahrain commercial banks.
- ✓ **GOSI & Finance GL Link:** Automated GOSI contribution tables and journal voucher generation for General Ledger posting.

4

Milestone 7.4: EOSB Gratuity Engine, Employee Self-Service (ESS) & Production Go-Live

Weeks 13 - 16
(4 Wks)

Delivery of Article 116 End of Service Benefits (EOSB) settlement engine, mobile-responsive ESS portal for staff requests and payslips, user acceptance testing, data migration, and full production cutover.

- ✓ **Article 116 Gratuity:** Precise EOSB indemnity computation based on service tenure, resignation laws, and leave encashment.
- ✓ **Mobile ESS Portal:** Self-service portal for leave applications, digital payslip downloads, and certificate requests.
- ✓ **Go-Live & UAT Sign-Off:** Full staff biometric onboarding, historical leave balance migration, and production deployment.

Section
5.0

Milestone Payment Structure & Invoicing Schedule (16-Week Model)

Commercial Billing
Governance

Billing is structured across 4 sequential verification gates tied directly to verifiable deliverable sign-offs:

GATE	MILESTONE DELIVERABLE FOCUS	TIMELINE	% FEE	INVOICE AMOUNT (BD)	VERIFICATION CRITERIA
Gate 7.1	Employee 360 Master, Document Dossier & Biometrics	End of Wk 4	25.00%	BD 1,523.637	CPR expiry alerts active & multi-branch ZKTeco biometric sync verified
Gate 7.2	Leave Accruals, Shift Rosters & Overtime Matrix	End of Wk 8	25.00%	BD 1,523.637	30-day leave accrual test run & overtime rules validated by HR
Gate 7.3	Bahrain WPS Payroll, GOSI Engine & Bank Export	End of Wk 12	25.00%	BD 1,523.637	Mock WPS CBB bank file verified & GL payroll voucher posted
Gate 7.4	EOSB Gratuity Engine, ESS Portal & Go-Live Acceptance	End of Wk 16	25.00%	BD 1,523.637	Art. 116 gratuity test approved & final production sign-off
Total Module 7 Investment (Fixed Deliverable Contract)			100.00%	BD 6,094.548	Complete Production Delivery

Section 6.0

Working Calendar, SLA, 15-Day Grace Period & Legal Terms

Master Service Level Agreement

This milestone schedule is governed strictly under the overarching Master Payment Governance & Service Level Framework (SL-POP-ERP-PAY-001);

Standard Working Calendar & Effort Accounting

All engineering effort calculations are structured upon a standard 5-day working week (Sunday through Thursday), 8 working hours per day (40 hours/week, 160 hours/month). Kingdom of Bahrain statutory public holidays are observed.

15-Day Verification Window & Deemed Acceptance

Upon milestone deployment notification, the Client is granted a 15-calendar-day UAT verification period. If no material defect is logged in writing within 15 days, milestone acceptance and invoice authorization occur automatically.

90-Day Production Bug-Fix Warranty

A comprehensive 90-calendar-day post-go-live warranty covers all functional discrepancies, calculation anomalies, and system defects originating from the agreed DOC-007 specification without additional charge.

2-Year Resource Rate Protection Guarantee

Engineering resource rates defined in Section 3.0 remain legally fixed for twenty-four (24) months from contract execution, ensuring predictable costs for any post-implementation change requests.

**Section
7.0**

Stakeholder Authorization & Digital Sign-Off Console

**Cryptographic Audit
Trail**

By executing digital signatures below, all authorized representatives formally validate the functional architecture, 16-week delivery roadmap, resource allocation pricing, and invoicing governance defined in this document:

FIRST PARTY – CLIENT / END-USER

Popular Auto Spare & A/C Parts Co. W.L.L.

Awaiting Client Signature

Signatory: Authorized Executive Director

Status: Pending Digital Execution

SECOND PARTY – LEAD IMPLEMENTATION PARTNER

SaNDS Lab Software Solutions

Awaiting SaNDS Lab Signature

Signatory: Lead Solutions Architect

Status: Pending Digital Execution

THIRD PARTY – ADVISORY & ASSURANCE PARTNER

UniGlobal Business Solutions

Awaiting UniGlobal Signature

Signatory: Principal Consultant

Status: Pending Digital Execution